

Towards Responsible Recommendations: A Daily Updated Ranking Model for Content Issue Detection

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Abstract

Most existing content issue detection models are built upon Multimodal Large Language Models (MLLMs). Although such approaches achieve high accuracy, they often suffer from poor timeliness, limited feature richness, and high serving costs: MLLM-based models are typically updated only monthly or quarterly and rely mainly on intrinsic video features (e.g., sampled frames and captions), underutilizing post-hoc user feedback and aggregated video and author-side signals. To address these challenges, we propose a daily updated ranking model for Content Issue Detection based on a Cross-Order Representation Fusion (CORF) architecture. The model supports day-level updates and integrates user behavior features, video- and author-side information, and MLLM-generated scores as inputs. Compared with MLLM-based approaches, our model is significantly smaller and can efficiently score all videos on a daily basis. Experiments on the *Adult-Content* category show higher recall at the same level of precision, and online A/B tests further demonstrate reduced user exposure to such content issues, contributing to a safer and more positive viewing experience.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence; Ranking; • Information systems** → **Recommender systems.**

Keywords

Responsible Recommendation, Content Issue Detection, Daily Updated Model, Ranking Model

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1 Introduction

In recent years, Multimodal Large Language Models (MLLMs)[1, 4] have achieved remarkable progress in unifying vision and language understanding. By combining visual encoders with large-scale language models and leveraging instruction tuning and chain-of-thought (CoT) reasoning[16], MLLMs demonstrate strong generalization, superior accuracy, and enhanced interpretability across a wide range of multimodal tasks. Their ability to jointly model textual and visual semantics has also driven substantial progress in content understanding and moderation, enabling automatic identification of inappropriate or policy-violating videos in recommendation feeds. As a result, MLLM-based methods[9, 14, 15] have become the dominant paradigm for content issue detection in large-scale recommendation systems.

Despite these strengths, MLLM-based methods face key challenges in real-world moderation. First, large-scale fine-tuning and infrequent update cycles (often monthly or quarterly) hinder adaptation to emerging issues and adversarial behaviors. Second, inputs are typically limited to intrinsic video features (e.g., frames and captions), underutilizing valuable user feedback signals—such as likes, dislikes, reports, and survey responses[17]—as well as author-level attributes that provide strong post-hoc cues. Third, the high computational cost constrains serving capacity: real-time scoring at scale is infeasible, so only a subset of high-traffic videos is evaluated, delaying the detection of newly uploaded harmful content.

To overcome these limitations, we propose a Daily Updated Ranking Model for content issue detection, which enables lightweight, timely, and comprehensive detection across all videos. The model supports day-level updates using newly collected user feedback and integrates multiple sources of information, including user behavior features, author attributes, and MLLM generated scores, into a unified ranking framework. As illustrated in Figure 1, the system comprises two main components: (1) a training part that constructs daily updated human-labeled samples, subsequently extracts video, author, and user-behavior features, and uses them together with the MLLM prediction score as inputs to the ranking model; and (2) a Serving Part that performs real-time scoring on every user–video pair to produce the final predict score. This design effectively addresses the issues of timeliness, feature richness, and serving scalability, offering a practical and responsible solution for large-scale content issue detection in recommendation systems.

To fully leverage the model’s input information and enable effective feature interactions, we propose a novel Cross-Order Representation Fusion (CORF) architecture, which captures and integrates both high-order and low-order feature interactions. We first validate our approach on the governance of *Adult-Content*, as defined in the TikTok Community Guidelines¹. In offline evaluations, our CORF-based daily updated ranking model achieves higher recall at the same precision level. Online A/B tests further show that the model improves the detection of *Adult-Content* issues, contributing to a safer and more positive viewing experience without negatively affecting overall engagement. Although our primary results focus on *Adult-Content* issues, additional evaluations indicate that the proposed framework generalizes well to other issue types, including *Clickbait* and *Promotional Content*.

2 Related Works

2.1 Multimodal Large Language Models

Multimodal Large Language Models (MLLMs) combine visual encoders with large language models for unified vision–language understanding. LLaVA [4, 7] introduces visual instruction tuning to align LLMs with vision encoders, showing strong generalization and reasoning across multimodal tasks. Building on this paradigm, the Qwen-VL series [1, 11] employs a multilingual LLM backbone and multi-stage training to unify visual grounding, instruction following, and conversational reasoning, achieving high efficiency and adaptability in real-world applications. Meanwhile, lightweight designs such as BLIP-2[5] and LLaMA-Adapter[19] use parameter-efficient fine-tuning and adapter-based integration to reduce serving costs while maintaining competitive performance.

2.2 Recommendation Models

Recommendation models have evolved significantly to capture complex feature interactions. Pioneering approaches like DeepFM [3] and xDeepFM [6] combine Factorization Machines for low-order interactions with DNNs for high-order patterns. Similarly, DCN [12] focuses on explicitly modeling bounded-degree feature crosses, while AutoInt [8] utilizes multi-head attention for high-order combinations. More recently, Mixture-of-Experts (MoE) architectures, such as HoME [13, 18], have emerged to enhance expressiveness and adaptability by dynamically routing inputs to specialized experts. However, despite these advancements, effectively fusing representations of different orders to fully exploit their complementary strengths remains an open challenge.

3 Daily Updated Ranking Model Structure

We propose a novel Cross-Order Representation Fusion (CORF) architecture to effectively capture and integrate both high-order and low-order feature interactions. The overall CORF architecture is illustrated in Figure 2. We first divide the input embedding into a sequence of n tokens, $\{I_0, I_1, \dots, I_{n-1}\}$, which form the input matrix $\mathbf{I} \in \mathbb{R}^{n \times D}$, where n is the sequence length and D is the dimension of each token. The core CORF module consists of a stack of CORF layers. Each layer is designed to extract and fuse feature interactions of varying complexity. Specifically, in each layer, the input

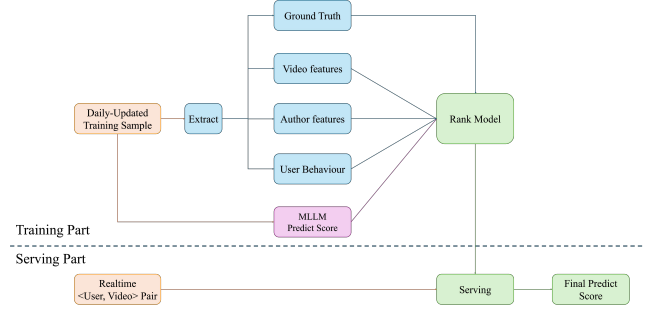


Figure 1: The framework of our proposed Daily Updated Ranking Model.

tokens $\mathbf{I}$ are processed in parallel by two distinct components: a High-Order Block that captures complex, high-order feature interactions, and a Low-Order Block that models simpler, low-order relationships. The resulting representations are then concatenated to produce the layer’s output tokens, $\mathbf{O}$. This architecture allows the model to progressively build richer representations by combining complementary feature information at each layer.

3.1 High-Order Block

Specifically, the High-Order Block first employs a scaled dot-product attention mechanism [10] to capture complex, high-order relationships among the input tokens. In this mechanism, the input tokens $\mathbf{I}$ are projected into queries ($\mathbf{Q}$), keys ($\mathbf{K}$), and values ($\mathbf{V}$). The output is a weighted sum of the values, where the weight for each value is determined by the dot-product similarity of its query and key. The operations of the High-Order Block for layer i are formally defined as:

$$\mathbf{H} = \text{Attention}(\mathbf{I}\mathbf{W}_j^Q, \mathbf{I}\mathbf{W}_j^K, \mathbf{I}\mathbf{W}_j^V) \quad (1)$$

Here, $\mathbf{H}$ denotes the resulting high-order token representations. The matrices $\mathbf{W}^Q \in \mathbb{R}^{D \times d_q}$, $\mathbf{W}^K \in \mathbb{R}^{D \times d_k}$, and $\mathbf{W}^V \in \mathbb{R}^{D \times d_v}$ are learnable projection parameters for the query, key, and value in the attention head. The input token dimension is D , while d_q , d_k , and d_v denote the dimensionalities of the queries, keys, and values, respectively. Following standard practice in attention mechanisms, we set $d_q = d_k$.

Subsequently, the high-order token representations are passed through a parameter-efficient, per-token SwiGLU layer. The PerTokenSwiGLU module applies an independent SwiGLU function with unshared weights to each token position, enhancing the model’s capacity to learn token-specific transformations. To further reduce the number of parameters, each PerTokenSwiGLU adopts a bottleneck architecture. The operation for the i -th token $\mathbf{T}_i$ is defined as:

$$\text{PerTokenSwiGLU}(\mathbf{T}_i) = \text{Swish}_1(\mathbf{T}_i\mathbf{W}_{1,i} + \mathbf{b}_{1,i})\mathbf{W}_{2,i} + \mathbf{b}_{2,i} \quad (2)$$

where the weights $\mathbf{W}_{1,i} \in \mathbb{R}^{D \times d}$ and $\mathbf{W}_{2,i} \in \mathbb{R}^{d \times D}$ are specific to token position i , with the intermediate dimension d being smaller than D to create a parameter-efficient bottleneck.

The final high-order block output, denoted as $\mathbf{H}^{\text{proc}}$, is computed as:

$$\mathbf{H}_i^{\text{proc}} = \text{PerTokenSwiGLU}(\mathbf{H}_i), \quad i = 1, \dots, n \quad (3)$$

¹<https://www.tiktok.com/community-guidelines/en-GB>

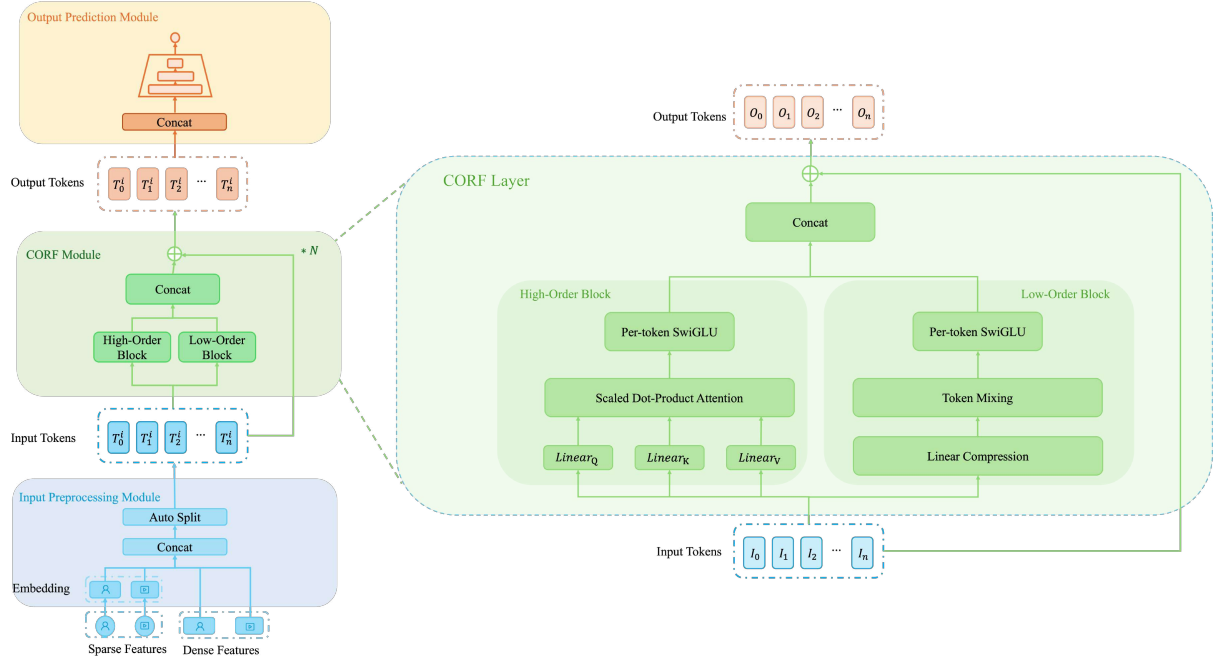


Figure 2: An overview of our proposed CORF architecture. It consists of an Input Preprocessing module, a stack of N CORF layers, and an Output Prediction module. Each CORF layer contains parallel High-Order and Low-Order blocks capturing interactions of different complexities. Their outputs are concatenated and fused with the input through a residual connection.

3.2 Low-Order Block

In contrast, the Low-Order Block is designed to model simpler, low-order feature interactions with high efficiency. As illustrated in Figure 2, it operates through a three-step process: Linear Compression, Token Mixing, and Per-token SwiGLU.

First, to reduce computational cost, a linear compression layer projects the input tokens $\mathbf{I} \in \mathbb{R}^{n \times D}$ to a shorter sequence of tokens $\mathbf{I}_{\text{comp}} \in \mathbb{R}^{n' \times D}$, where $n' < n$. This is achieved by multiplying the input tokens with a learnable projection matrix $\mathbf{P} \in \mathbb{R}^{n' \times n}$. The compressed tokens are then processed by a Token Mixing mechanism, a technique adapted from RankMixer [20], to facilitate information exchange across tokens in a computationally efficient manner while enhancing inter-token diversity. The operations are formally defined as:

$$\mathbf{L} = \text{TokenMixing}(\mathbf{I}_{\text{comp}}) \quad (4)$$

Here, $\mathbf{L}$ represents the resulting low-order token representations.

Similar to the high-order block, the low-order token representations are passed through a per-token SwiGLU layer to obtain the final low-order block output $\mathbf{L}^{\text{proc}}$, as defined by the following equation:

$$\mathbf{L}_i^{\text{proc}} = \text{PerTokenSwiGLU}(\mathbf{L}_i) \quad i = 1, \dots, n' \quad (5)$$

The final step within each CORF layer is to concatenate the computed high-order $\mathbf{H}^{\text{proc}}$ and low-order $\mathbf{L}^{\text{proc}}$ representations along the token dimension, followed by integrating the result with the layer's original input through a residual connection.

4 Experiment

4.1 Experimental Setup

4.1.1 Training&Test Datasets. We first apply this method to *Adult-Content* issue governance. We use 4 million videos with human-reviewed labels to fine-tune the MLLM and obtain its predicted scores. These data are then used to train our Daily Updated Rank Model. Following initialization, the Rank Model is updated daily with approximately 200K new samples added each day. For evaluation, we construct a test set by re-sampling 20K videos, ensuring no overlap with the training data.

4.1.2 Evaluation Metrics. We comprehensively evaluate our model through both offline and online experiments. In the offline evaluation, we examine recall at different precision levels alongside model size to assess the balance between performance and efficiency. In the online A/B testing, we further validate the model's real-world effectiveness using core metrics such as user Stay Duration (SD) and Ecosystem Negative View Rate (ENVR), which respectively measure user engagement and the proportion of views associated with undesirable or low-quality content.

4.1.3 Implementation Details. Our proposed model is implemented in TensorFlow with two CORF layers to balance effectiveness and real-time serving requirements for user-video pairs in TikTok's *For You* feed. We use RMSProp with a learning rate of 0.01 and a batch size of 256, training on 8 NVIDIA A100 GPUs. In production, the model is updated daily and deployed on approximately 150 NVIDIA A100 GPUs to maintain freshness at scale.

Table 1: Comparison of our CORF-based daily updated rank model with MLLM-based models in terms of recall rate at different precision levels.

	Params (M)	Recall Rate		
		@P30	@P40	@P50
VLMO-Base	175	22.82%	13.64%	10.82%
Qwen2.5-0.5B	500	26.12%	19.53%	14.59%
CORF (ours)	1.8	29.41%	22.35%	17.17%

Table 2: Online A/B test results comparing our CORF-based daily updated rank model with the MLLM-based baseline on *Adult-Content* issue governance.

	SD	ENVR	ENVR@Adult-Content
Qwen2.5-0.5B	-0.051%	-0.253%	-2.873%
CORF (ours)	-0.047%	-0.402%	-3.593%

4.2 Comparisons with MLLM-based Models

We compare our proposed CORF-based daily updated rank model with two lightweight MLLM-based backbones, *VLMO-Base* [2] and *Qwen2.5-0.5B* [1], on the *Adult-Content* issue category. Both were among the highest-performing open-source models available for this use case at the time. As shown in Table 1, our CORF-based model achieves the best overall performance while being orders of magnitude smaller. With only 1.8M parameters, it consistently surpasses the MLLM-based baselines across all precision levels, demonstrating strong representational capability, computational efficiency, and scalability for real-time, large-scale deployment. Moreover, its daily update mechanism enables rapid adaptation to emerging content patterns, ensuring sustained relevance and robustness in dynamic online environments.

4.3 Online AB Test Results

Table 2 presents the online A/B test results for *Adult-Content* issue governance, comparing our CORF-based daily updated rank model with the MLLM-based baseline *Qwen2.5-0.5B*. Our model achieves a lower SD loss (-0.047% vs. -0.051%), indicating a smaller impact on user watch time while maintaining ranking effectiveness. More importantly, it delivers substantially larger reductions in ENVR (-0.402% vs. -0.253%) and ENVR@Adult-Content (-3.593% vs. -2.873%). These improvements demonstrate the model’s enhanced ability to surface *Adult-Content* issue while minimizing negative effects on user engagement. Overall, the results highlight the effectiveness of the CORF architecture and the daily update mechanism for real-time, large-scale content quality governance.

5 Fairness Discussion

By incorporating posterior user behavior, our approach reduces inadvertent misclassification compared to MLLM-based models. Recognizing that content perception is inherently subjective, future iterations will incorporate user-level demographics (e.g., age, gender, and region) and integrate survey-based labels to further enhance user-centric fairness.

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